PROSTHETIC ARM WITH ADAPTIVE GRASP MODES

ECD416 Project Phase II: Report Submitted by

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Towards the partial fulfillment of the requirement for the award of B. Tech. degree in

Electronics and Communication Engineering

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March 2025

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CERTIFICATE

Certified that this is a bonafide report of the project work done by **ASHWIN ANOOPDAS (Reg. No. KNR21EC024)**, **FATHIMA PALOT (Reg. No. KNR21EC035)** , **GOURAV K. (Reg. No. KNR21EC042)** and **ULLAS RAVI (Reg. No. KNR21EC098)** on the topic **PROSTHETIC ARM WITH ADAPTIVE GRASP MODES**, towards the partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Communication Engineering under APJ Abdul Kalam Technological University during the academic year 2024-2025.

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DECLARATION

We, the undersigned, hereby solemnly declare that this project report titled **PROSTHETIC ARM WITH ADAPTIVE GRASP MODES**, submitted for the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering from APJ Abdul Kalam Technological University is a bonafide record of our own work carried out under the supervision of **Dr. SAJITH K.**.

Wherever we have used materials (data, theoretical analysis, and text) from other sources, we have adequately and accurately cited the original sources.

We also declare that this work has not been submitted to any other institution in this University or any other University.

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ACKNOWLEDGMENTS

We would like to take this opportunity to extend our sincere thanks to people who contributed to the progress of this project. Our work will be incomplete without mentioning all the people who helped us to make it real.

We are extremely grateful to the Principal, **Dr. RAJESH K. N.**, Govt. College of Engineering, Kannur, for providing the necessary facilities. We would like to express our sincere gratitude to **Dr. A. RANJITH RAM**, Head of Department of Electronics and Communications for giving us all the support and confidence to work with our project. We express our sincere gratitude to our project guide, **Dr. SAJITH K.** and project co- ordinator, **Prof. ROHIT P.** for their great support.

We would like to thank all our teaching and non-teaching staff of Govt. College of Engineering, Kannur for their valuable help in advancing this phase of our project . We extend our thanks to family members and especially to our friends whose encouragement and assistance during challenging times have greatly contributed to our project. We also wish to thank our peers who helped us in performing certain tests and gave us various suggestions.

ABSTRACT

The **PROSTHETIC ARM WITH ADAPTIVE GRASP MODES** is designed to empower individuals with limb loss by enhancing their ability to manage everyday tasks independently, specifically within a dining environment. The system combines vision-based object recognition and dynamic grip control to facilitate safe and efficient interaction with common dining items.

A camera captures real-time images that are processed to identify tableware such as cups, bowls and spoons, and provisions like apples. Based on this visual data, the system adapts the prosthetic arm's grasp, depending on the type of object. Force sensors continuously provide feedback on grip strength, allowing the prosthetic arm to adjust its pressure dynamically, ensuring secure handling without risk of damage.

This project presents a functional and accessible prosthetic solution that combines advanced image processing and sensor feedback to restore autonomy in daily dining tasks for individuals with upper limb loss.

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LIST OF ABBREVIATIONS

ADC Analog-to-Digital Converter

CNN Convolutional Neural Networks

EMG Electromyography

FSR Force Sensing Resistor

GPIO General Purpose Input/Output

GPU Graphics Processing Unit

I2C Inter-Integrated Circuit

NumPy Numerical Python

OpenCV Open Source Computer Vision Library

PLA Polylactic Acid

PWM Pulse Width Modulation

SPI Serial Peripheral Interface

YOLO You Only Look Once

CHAPTER 1

INTRODUCTION

Prosthetic technology has long played a crucial role in improving the lives of individuals with limb loss or disability, enabling them to regain functionality, confidence, and independence in daily tasks. However, traditional prosthetic devices often lack the ability to adapt to real-life situations, such as distinguishing between delicate and firm objects or responding autonomously to the demands of different environments. These limitations can make daily activities, such as dining, challenging and cumbersome, often requiring assistance from caregivers.

The Prosthetic Arm with Adaptive Grasp Modes is designed to address these challenges by integrating intelligent, real-time control into the prosthetic device. Focusing on common dining tasks, this prosthetic arm utilizes a vision-based recognition system and force feedback to adapt its grip to specific objects, like tableware and food provisons.

This approach reduces the physical and cognitive load on users, allowing for a more natural and intuitive experience while handling dining objects. By combining image processing, adaptive grip control, and force sensor feedback, this project aims to provide a highly functional and accessible prosthetic solution, empowering individuals with disabilities to achieve greater independence and restore autonomy in their everyday lives.

1.1 Objectives

1. Design a prosthetic arm with grip modes that adjust dynamically based on the object being handled.
2. Implement real-time object recognition, enabling safe, efficient handling of objects with minimal user intervention.
3. Utilize force sensors to monitor grip pressure continuously, enabling the arm to adjust grip strength dynamically to prevent spills or damage.

1.2 Motivation

1. Individuals with limb loss often face challenges in performing basic tasks, such as dining, independently. This project seeks to empower users by providing a prosthetic arm that autonomously adapts to different dining scenarios.
2. Many prosthetics lack adaptive grip control, which limits their functionality in real-life situations. Introducing adaptive grasp modes improves usability and effectiveness in daily tasks.
3. By automating grip adjustments through object recognition, the system minimizes user effort and reduces mental strain during tasks that require fine motor skills.
4. Developing a cost-effective, adaptable, and reliable prosthetic solution helps make advanced assistive technology accessible to a broader population.

CHAPTER 2

LITERATURE SURVEY

The Johns Hopkins Applied Physics Laboratory [1] has created various prosthetic limbs and technologies, including the Modular Prosthetic Limb, an anthropomorphic prosthetic arm which can be customized to meet the specific needs of a patient's injury. Developed as part of the Revolutionizing Prosthetics program, it aims to restore both motor and sensory capabilities to upper-limb amputees. It features an interface for closed-loop control, high-resolution tactile and position sensing, and offers human-like strength and dexterity.

By employing various image processing algorithms, such as image segmentation and morphological transformations, the robotic arm can accurately determine the position of objects, leading to improved efficiency in its movements and operations. This integration of computer vision, presented by Seiffert et al. [2] , not only reduces computational costs but also minimizes energy consumption. Varol et al. [3] proposed a depth camera-based intent recognition system that can effectively identify various activity modes (standing, walking, running, stair ascent, and stair descent) for robotic prosthetic legs, with a high accuracy of 99%. Martin et al. [4] put forward the semi-autonomous operation mode facilitated by the computer vision algorithm, which simplifies the control process while allowing for simultaneous multi-joint movement of the prosthesis.

Therese et al. [5] presented the development of a robotic arm equipped with real-time image processing capabilities, utilizing a Raspberry Pi for both automated and manual operation. The project aims to create a robot capable of detecting and placing specified objects. Ultimately, this system allows for both manual interven-

tion and fully automated task execution, showcasing the potential of Raspberry Pi in robotic applications.

Narote et al. [6] introduced a mechanically active and multifunctional 3-D printed prosthetic arm designed to physically and functionally replicate the capabilities of a natural human hand. The project successfully achieves the goal of enabling a range of actions and activities akin to those performed by a typical human hand. The prototype is designed as a low-cost model, ensuring that it can provide effective assistance to users of prosthetic devices.

In the work presented by Agcayazi et al. [7] , the design and results of a novel limb-prosthetic interface force sensor that is engineered to respond to both shear and normal forces simultaneously, has been introduced. Gaetani et al. [8] developed an electronic system for a prosthetic limb that is controlled by a wireless myoelectric armband. This prosthesis is designed to provide users with feedback about their surroundings, including object temperature and the grip force exerted by the fingers on grasped items.

Two autonomous grasping tasks utilizing force feedback control have been successfully demonstrated on the Modular Prosthetic Limb by McGee et al. [9]. These tasks enhance the functionality of the robotic arm, complementing its primary role as an advanced prosthetic limb.

CHAPTER 3

PROPOSED SYSTEM

3.1 Problem Definition

Individuals with upper limb loss encounter significant challenges in performing daily tasks, especially those requiring fine motor skills and delicate handling, such as dining. Traditional prosthetics generally operate on simple, fixed-grip mechanisms that require the user to manually adjust grip strength or position. This rigidity makes it difficult for users to handle various objects with differing fragility, weight, or content. As a result, users often face unintentional breakage, spills, or slippage, increasing their dependency on caregivers for basic tasks.

Furthermore, current prosthetic solutions typically do not incorporate real-time feedback or environmental awareness, forcing users to rely on manual adjustments and increasing their cognitive load. The lack of dynamic sensors, adaptive algorithms, and real-time object recognition in these prosthetics means they cannot autonomously adjust to the demands of specific tasks.

The aim is to create a system capable of recognizing different dining objects and adjusting its grip strength and positioning automatically, thus empowering users to handle items safely and confidently, and ultimately enhancing their independence and quality of life.

3.2 Essential Python Libraries

In this project, Python libraries play a crucial role in building and operating the prosthetic arm's capabilities, which includes Ultralytics YOLO for object detection, OpenCV for video processing, Adafruit ServoKit for servo control, and NumPy for numerical operations. Together, these libraries form a powerful toolkit, enabling real-time detection, adaptive control, and robust data processing for the prosthetic arm system.

3.2.1 Ultralytics YOLO

The Ultralytics YOLO library is a state-of-the-art implementation of the YOLO object detection algorithm. It is designed for fast and accurate object detection in images and video streams. This library is used to load a custom-trained YOLO model and detect objects like the prosthetic, apple, cup, etc. The library provides an easy-to-use interface for processing detection results, including bounding box coordinates, class labels, and confidence scores.

3.2.2 OpenCV

OpenCV is a powerful library for real-time computer vision tasks. In the project, it is used to capture video frames from a camera and process them for object detection. OpenCV provides functions for displaying video streams, drawing bounding boxes around detected objects, and overlaying text on the frames. OpenCV is essential for integrating the camera feed with the object detection pipeline and visualizing the results in real-time.

3.2.3 NumPy

NumPy is a fundamental library for numerical computing in Python. It provides support for arrays, matrices, and a wide range of mathematical operations. NumPy is used to handle the numerical data generated during object detection, such

as bounding box coordinates and distances between objects. It is particularly useful for performing vectorized operations, which are faster and more efficient than traditional loops.

3.2.4 Adafruit ServoKit

The Adafruit ServoKit library is used to control servo motors via the PCA9685 servo driver. It provides an easy-to-use interface for managing multiple servos connected to the PCA9685, to set precise angles and control their movement. The library supports setting custom pulse width ranges, which is essential for ensuring compatibility with different servo models like the MG995.

3.2.5 spidev

The spidev library is used to communicate with SPI (Serial Peripheral Interface) devices, such as the MCP3008 ADC. It is used to read analog values from the FSR402 force-sensitive resistor. The library allows for the sending and receiving of data over the SPI bus, enabling precise measurement of the force applied to the sensor.

3.2.6 time

The time library is a built-in Python module that provides functions for handling time-related tasks. In the project, it is used to add delays between servo movements, control the timing of the object detection loop, and synchronize various operations. It is also used to control the frequency of object detection and servo activation, ensuring that the system operates efficiently without overloading the Raspberry Pi.

3.3 Algorithm and Block Diagram

To enable the prosthetic arm to interact with dining objects autonomously and adaptively, a multi-step algorithm is implemented. This algorithm integrates real-time image processing, distance sensing, and force feedback to detect objects, assess their contents, and apply the appropriate grip strength for safe handling.

1. Power on the prosthetic arm, activating the Raspberry Pi, webcam, servo motors and force sensors.
2. Capture real-time images of the dining environment using the webcam and OpenCV.
3. Detect objects in the captured images using the pre-trained YOLO model.
4. Calculate the distance between the prosthetic arm and each detected object using the bounding box edges.
5. Actuate the servo motors to move the fingers and wrist of the prosthetic arm to grasp the detected object.
6. Engage the force sensor on the prosthetic to monitor grip pressure in real time. Continuously process the sensor data to adjust the servo angles and prevent over-gripping or under-gripping.
7. After the object has been successfully handled or lifted, release it back on to the surface.

Fig. 3.1 illustrates the block diagram of the proposed system, which is centered around the Raspberry Pi, functioning as both the microcontroller and image processing unit. The system includes a camera module and force sensor as the input, along with a servo driver to control the servo motors in the prosthetic arm.

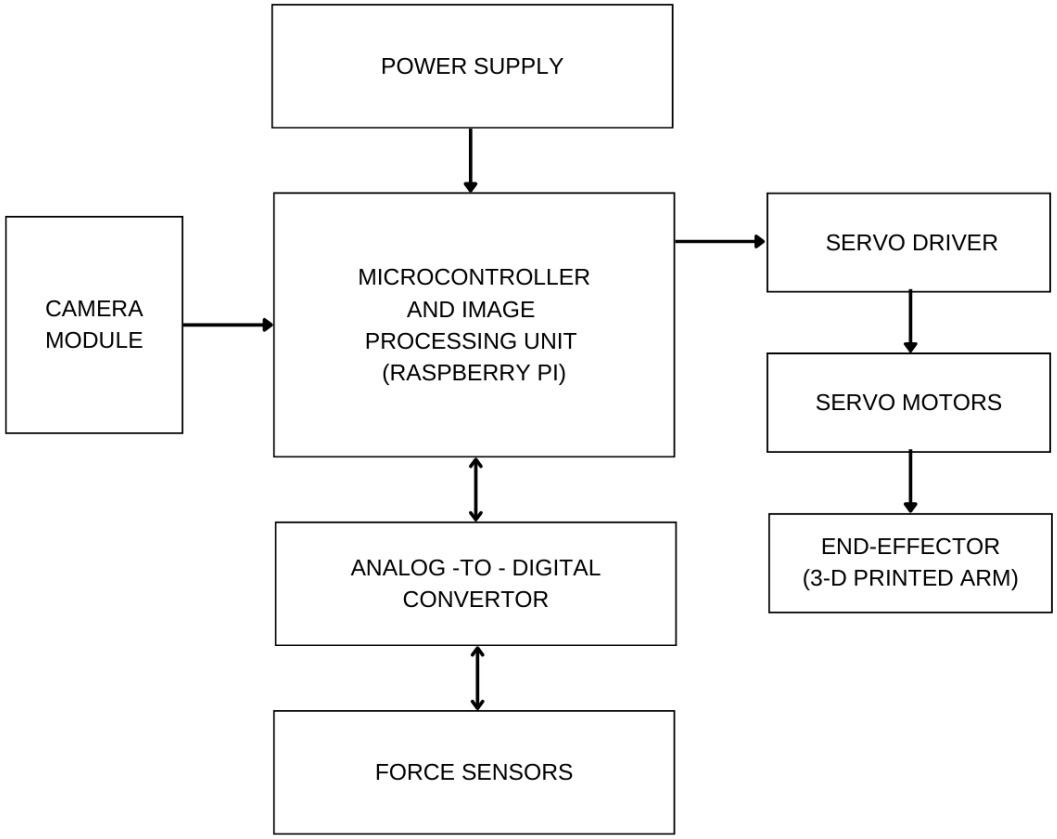

Fig. 3.1. Block Diagram of the Proposed System

3.4 3D Printed Prosthetic Arm

The InMoov hand and forearm shown in Fig. 3.2 showcases a detailed multi-joint robotic arm constructed entirely from 3D printed components. This design consists of 28 individual parts, all fabricated using PLA material [10].

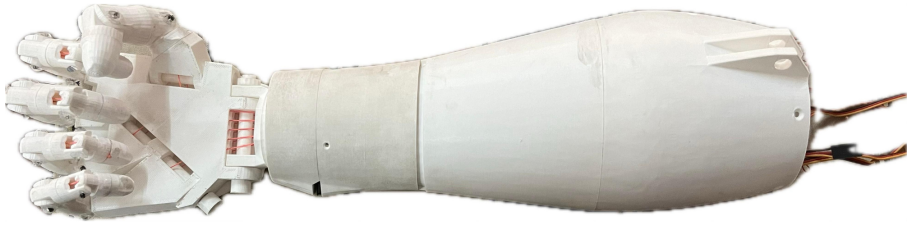

Fig. 3.2. Prosthetic Arm Model

Each finger has a degree of freedom for flexion and extension, allowing it to curl and straighten. Additionally, the wrist component offers 180 degrees of rotation, enabling a wide range of motion for more natural movements. To facilitate the articulation of the fingers and various movable joints throughout the prosthetic arm, nylon fishing line has been utilized as the tendon material.

3.5 Hardware Design

The hardware design of this project encompasses all essential components required for the functioning of the prosthetic arm, including the Raspberry Pi, webcam, servo driver and servo motors, force sensor and analog-to-digital convertor, as shown in Fig. 3.3.

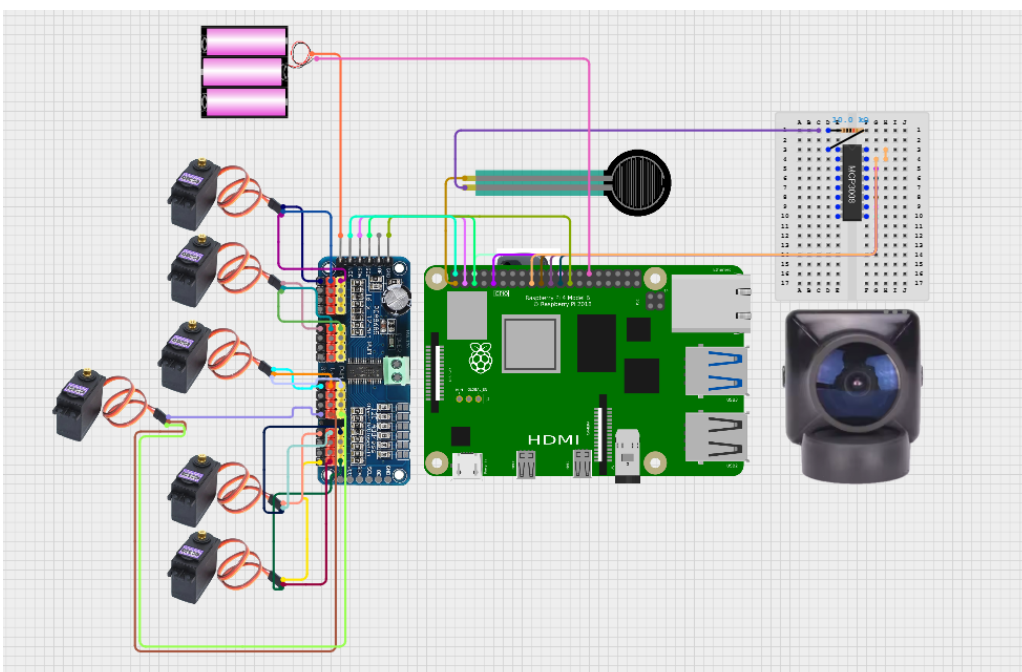

Fig. 3.3. Circuit Diagram of the Proposed System

3.5.1 Raspberry Pi and Webcam

The Raspberry Pi, as shown in Fig. 3.4. [11],serves as the primary processing unit for the prosthetic arm, coordinating data from sensors and controlling the motors. It runs image processing algorithms to detect the dining environment.

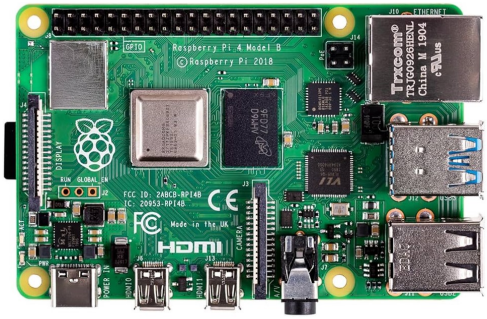

Fig. 3.4. Raspberry Pi 4 Model B

The webcam, as seen in Fig. 3.5. [12], mounted on a cap worn by the user, captures real-time video feeds, which are then analyzed by the Raspberry Pi to determine the presence and type of items on the table. By using Python libraries, the Raspberry Pi processes images to classify objects, enabling the prosthetic arm to adjust its grip accordingly.

Fig. 3.5. Webcam

3.5.2 Servo Motor Driver

The PCA9685, as shown in Fig. 3.6. [13], is a 16-channel PWM driver that allows microcontrollers like the Raspberry Pi to control multiple servos efficiently. Since the Raspberry Pi's GPIO pins can only generate limited PWM signals, the PCA9685 acts as an intermediary, using I2C communication to send precise control signals to up to 16 servo motors simultaneously.

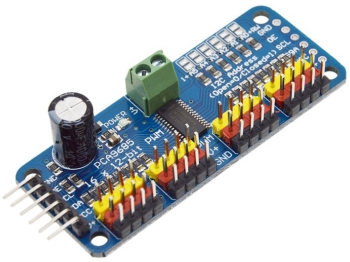

Fig. 3.6. PCA9685 Servo Motor Driver

3.5.3 Servo Motors

The MG995, as seen in Fig. 3.7. [14], is a high-torque digital servo motor widely used in various applications due to its strong metal gears and high durability. Its PWM-based position control allows for smooth and precise motion adjustments, making it suitable for applications that require gripping and wrist rotation. However, due to its high current draw, it is recommended to use an external power source with sufficient capacity when operating multiple MG995 servos.

Fig. 3.7. MG995 Servo Motor

3.5.4 Force Sensors

Force sensor, as represented in Fig. 3.8. [15], play a key role in adapting the prosthetic arm's grip to different objects. Mounted on one of the fingers, these sensors provide real-time feedback on the force being applied to an object. When gripping a fragile item, the sensors detect the pressure and signal the system to adjust so as to avoid deformation. Conversely, when handling a heavier item, the system

maintains a firmer grip to prevent slippage. This feedback loop, enabled by the force sensors, allows the prosthetic arm to mimic the nuanced touch sensitivity of a human hand, crucial for handling various dining items safely and effectively.

Fig. 3.8. FSR402 Force Sensor

3.5.5 Analog-to-Digital Converter

Force sensors produce analog signals based on the pressure applied, but the Raspberry Pi, a digital processor, requires these signals in digital form to process them accurately. The MCP3008 ADC, as shown in Fig. 3.9. [16], converts the analog output from the force sensors into digital signals that the Raspberry Pi can read. It operates over SPI communication, ensuring fast and efficient data transfer with minimal processing delay.

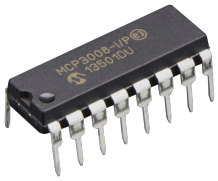

Fig. 3.9. MCP3008 ADC

3.6 Software Design

The prosthetic arm system integrates computer vision, real-time object detection, and adaptive control, enabling autonomous interaction with dining objects. The system is powered by a YOLO11-based deep learning model, which is trained to recognize the prosthetic arm, dining objects (such as spoons, cups, and apples), and humans. The trained model is deployed on a Raspberry Pi, where it processes the camera feed and detects the objects. The overall software workflow is illustrated in Fig. 3.10., which provides a detailed visual representation of the detection, processing, and control mechanisms within the system.

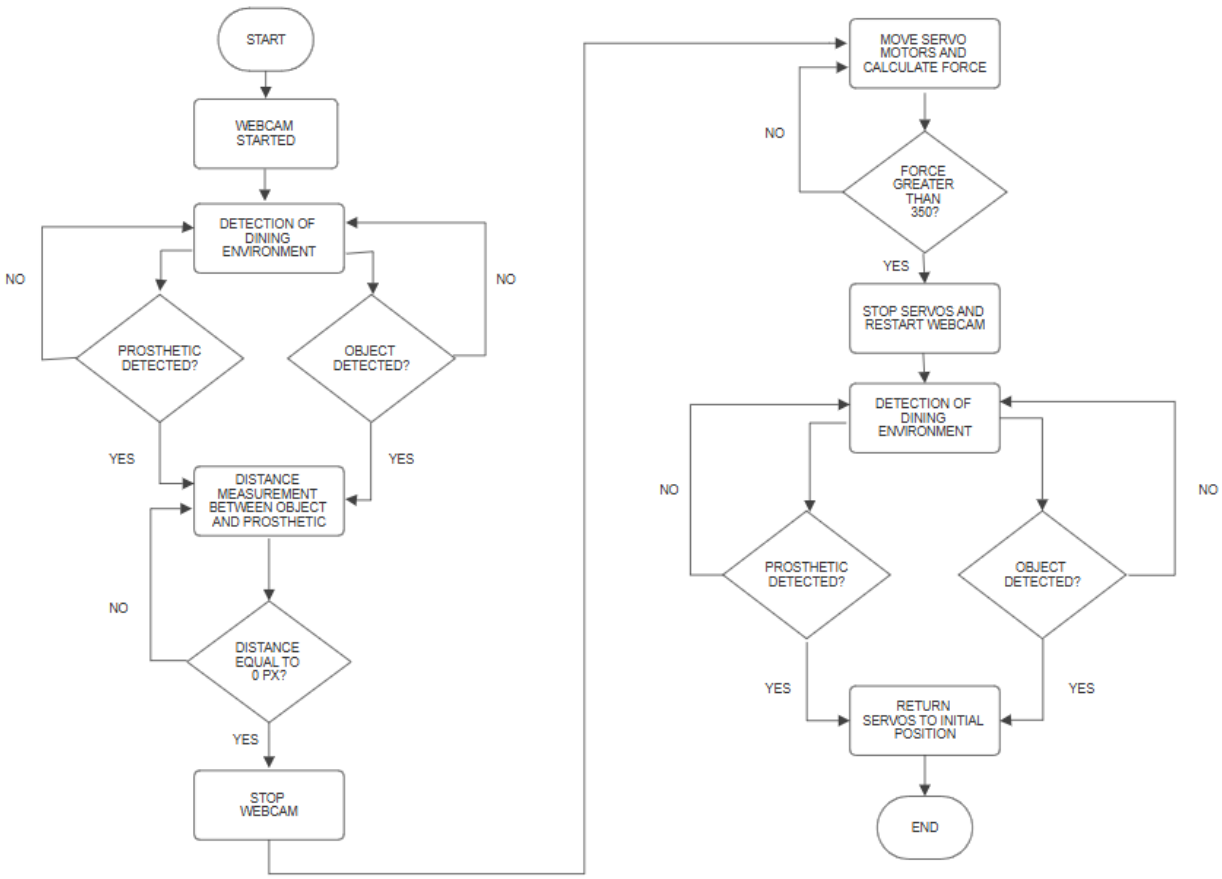

Fig. 3.10. Overall Workflow of the System

3.6.1 Label Studio

Label Studio is an open-source image annotation tool used to manually label and annotate objects in images. It supports YOLO formats, making it ideal for object

detection model training.

1. Install Label Studio using Python
2. Load images from the dataset directory.
3. Draw bounding boxes around objects like the prosthetic arm, cups, spoons, apples, and humans.
4. Assign class labels and save annotations in YOLO format.
5. Export labeled dataset for YOLO11 training.

3.6.2 YOLO11

YOLO is a family of real-time object detection machine-learning algorithms. Multiple machine-learning algorithms are used for object detection, one of which is CNNs. YOLO11 is a state-of-the-art object detection model known for its real-time performance and high accuracy. It was chosen due to its ability to detect multiple objects within a single frame with minimal latency.

The YOLO11 architecture was fine-tuned using the labeled dataset, and hyperparameters like batch size, learning rate, and number of epochs were optimized for better accuracy. After training, the best model weights (.pt file) were exported for deployment.

3.6.3 Google Colab

Since training deep learning models on a local machine is computationally intensive, Google Colab was used for model training. Colab provides free GPU acceleration, allowing for faster processing.

1. Upload dataset to Google Drive.
2. Install YOLO11 and required dependencies

3. Load and preprocess data using PyTorch.

4. Train YOLO11 on the dataset, adjusting parameters for optimal performance.

5. Evaluate model accuracy, and export the best model weights for deployment.

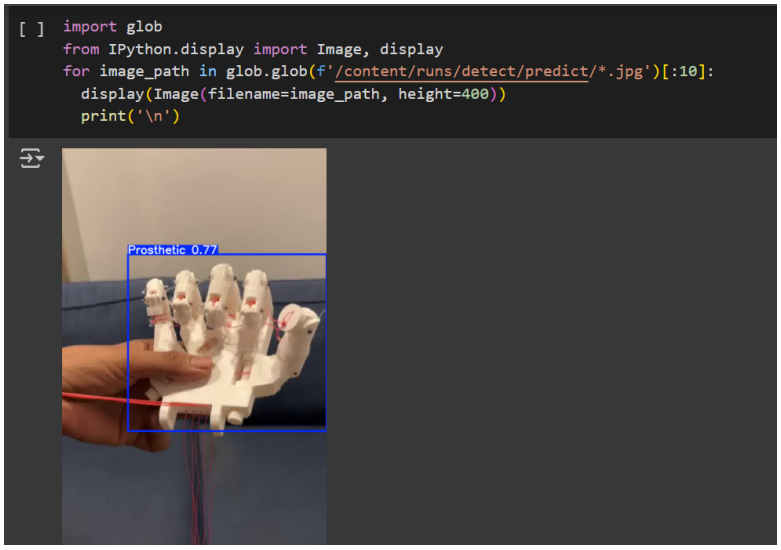

Fig. 3.11. YOLO model tested on Google Colab

3.6.4 Deployment on Raspberry Pi

Once the YOLO11 model was trained, it needed to be optimized for deployment on the Raspberry Pi. The original PyTorch model (.pt format) was too large, so it was converted to NCNN format, which is more efficient for ARM-based processors.

To enable real-time object detection, the optimized NCNN model was deployed on the Raspberry Pi. A Python script was created to capture video input, run object detection, and display bounding boxes on detected objects.

3.6.5 Distance Estimation

To enable the prosthetic arm to interact effectively with dining objects, it is essential to measure the distance between the prosthetic arm and detected objects. The system achieves this by analyzing bounding box coordinates from the YOLO11

model and calculating the distance between the prosthetic arm and target objects in real-time.

1. Each detected object is assigned a bounding box with coordinates in the form (xmin, ymin, xmax, ymax)
2. The system first checks for bounding box overlap. If no overlap, calculates minimum edge-to-edge distance using horizontal and vertical separation
3. The Euclidean distance between edges is then computed using NumPy.
4. When edge-to-edge distance reaches zero (physical contact detected), the servo motors activate to grasp the object.

3.6.6 Force Sensing and Grip Adjustment

1. Read Force Data from FSR via MCP3008. A predefined threshold is set for detecting excessive grip pressure.
2. If the force value remains within the threshold, the current grip is maintained.
3. If the force value exceeds the threshold, the system reduces the servo angles to loosen the grip.
4. If force drops suddenly, the system tightens the grip slightly.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Results

The prosthetic arm was assembled with six servo motors controlled via the PCA9685 driver. The FSR was placed on the middle finger to detect grasp pressure. The system was powered by a Raspberry Pi, which handled real-time processing and servo control. The hardware was successfully integrated, and all components functioned as expected.

Fig. 4.1. Final prosthetic arm prototype with exposed servo motor mechanism

The YOLO model was tested for real-time object detection, focusing on objects such as bottles, cups, and apples. The model successfully identified and labeled objects with bounding boxes. The system accurately calculated the distance between the prosthetic hand and detected objects using the bounding box edges. When edge-to-edge distance reaches zero, the servo motors were activated.

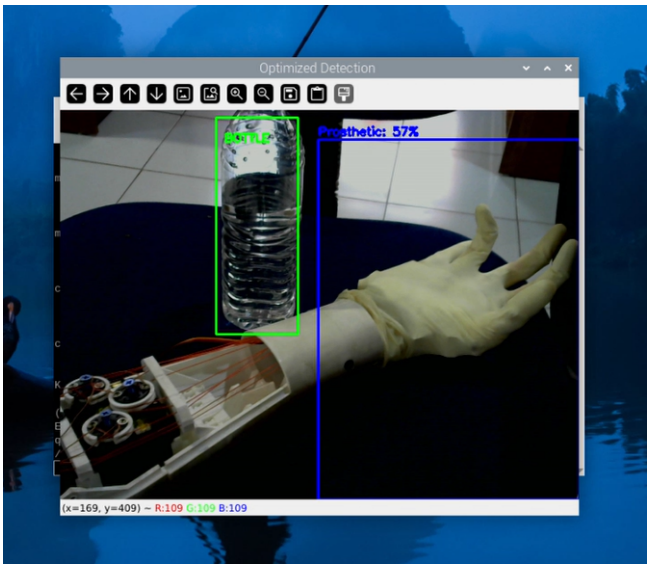

Fig. 4.2. Object detection system identifying both prosthetic arm and bottle

The FSR sensor effectively measured applied force, and when its value exceeded the threshold, servo movement stopped, ensuring a secure yet careful grasp.

(a.) (b.)

Fig. 4.3. Prosthetic arm handling (a) apple and (b) bottle, validating multi-object grasping capability

The fully integrated system successfully detected objects, measured distances, and controlled servo movements based on proximity and force feedback. A 0.5-second delay before activation improved stability. The slow servo motion allowed for a precise grasp, reducing sudden movements. The system effectively demonstrated object interaction, validating the design approach.

4.2 Discussions

The development and testing of the prosthetic arm system involved integrating multiple hardware and software components to achieve real-time object detection and grasping functionality. The results demonstrate the effectiveness of the system but also highlight certain challenges and areas for improvement.

1. The integration of the servo driver with six servos allowed for precise and controlled movement of the prosthetic fingers. The ADC effectively converted analog force sensor readings to digital values, ensuring responsive grip control. The Raspberry Pi successfully handled real-time computations, though minor delays were observed due to processing constraints.

2. The YOLO model successfully identified target objects such as bottles, cups, and apples, with a reasonable detection accuracy. The bounding box center calculations provided a simple yet effective method for estimating object distance. However, varying lighting conditions and occlusions sometimes led to false detections or misclassifications.

3. The system was programmed to activate the servos when an object came within a predefined threshold distance. The slow and gradual motion of the servos helped ensure a secure grip. However, minor jittering in servo motion was observed in some cases, which could be improved by optimizing the PWM signals.

4. The FSR provided a reliable means of detecting when an object was firmly grasped. When the force value exceeded the threshold, the servo motion stopped, preventing excessive pressure on the object. The response time of the sensor was satisfactory, but further tuning of sensitivity thresholds could enhance performance, particularly for delicate objects.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Prosthetic Arm with Adaptive Grasp Modes project demonstrates a significant advancement in the field of assistive technology by integrating real-time object recognition with adaptive grip control to support individuals with upper limb loss. By focusing on tasks within a dining environment, this system enables users to interact with everyday items, autonomously and effectively.

By combining computer vision, force sensing, and servo control, the prosthetic arm offers a functional and responsive solution for users. Through YOLO-based object detection and distance estimation, the arm is capable of recognizing and interacting with various objects, making it particularly useful in scenarios such as dining assistance. By providing automated grasping and adaptive feedback mechanisms, the prosthetic enhances user independence and usability, offering a practical alternative to expensive commercial solutions.

The project's success in creating a functional, intuitive system underscores the transformative potential of adaptive prosthetics, contributing to improved quality of life and greater independence for users.

5.2 Future Scope

The future scope of the Prosthetic Arm with Adaptive Grasp Modes is vast, with multiple opportunities for expanding and refining its capabilities. Improving the

robustness of the image processing algorithms to handle diverse lighting conditions and various object orientations would further enhance the system's reliability in real-world settings.

The integration of EMG sensors would allow users to control the prosthetic intuitively using muscle signals, bridging the gap between human intention and robotic movement. Additionally, incorporating haptic feedback systems could enhance user experience by providing sensory feedback, allowing individuals to perceive the texture, weight, or pressure of objects being held. Future iterations could also explore 3D-printing bio-compatible materials for enhanced comfort and wearability.

The use of machine learning algorithms could improve adaptive grasping strategies, enabling the prosthetic to learn from past interactions and optimize grip patterns for different objects. Real-time AI-based motion prediction could further refine responsiveness, making the arm behave more naturally. Wireless communication technologies could enable remote diagnostics, updates, and control, making the system more user-friendly and customizable.

Beyond dining, this technology could be adapted for a wide range of daily activities, such as personal care and household tasks, broadening its applicability and utility. These developments represent promising directions for future work, pushing the boundaries of prosthetic technology toward creating a highly functional, adaptable, and intelligent assistive device that fosters autonomy and inclusivity across multiple aspects of life.

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APPENDIX

DATASHEETS

1. **Raspberry Pi 4 Model B**

   Introduction

   The Raspberry Pi 4 Model B (Pi 4B) is the first of a new generation of Raspberry Pi computers, supporting more RAM and with significantly enhanced CPU, GPU, and I/O performance.

   Features

   - Quad core 64-bit ARM-Cortex A72 running at 1.5GHz
   - Bluetooth 5.0 with BLE
   - 1x SDCard
   - 2x micro-HDMI ports supporting dual displays up to 4Kp60 resolution
   - 1x Gigabit Ethernet port (supports PoE with add-on PoE HAT)
   - 2x USB2ports and 2x USB3ports
   - 1x Raspberry Pi camera port (2-lane MIPI CSI)
   - 28x user GPIO

   Physical Specifications

   - Dimensions: 85.6mm x 56.5mm
   - Weight: 46g
   - Power Requirement: 5V DC via USB-C connector (minimum 3A)

Operating System Support

- Officially supported operating systems: Raspberry Pi OS, Ubuntu, etc.
- Wide community support for various Linux distributions

Usage Recommendations

- Use a suitable power supply to ensure stable operation
- Properly configure and update the operating system for security
- Use a heat sink or cooling solution for heavy processing tasks
- Properly shutdown the system to avoid data corruption

Documentation and Support

- Official documentation and resources available on the Raspberry Pi Foundation's website.
- Dedicated forums and community support.

2. **PCA9685 Servo Motor Driver**

Introduction

The PCA9685 16-Channel PWM Servo Driver is an I2C-based PWM controller designed for efficient servo motor control. It allows microcontrollers to control up to 16 servo motors simultaneously.

Features

- 16 independent PWM output channels for servo motors or LEDs.
- I2C communication (2-wire interface) with configurable addresses, allowing multiple boards to be chained.
- 12-bit resolution (4096 steps per pulse cycle) for precise motor control.
- **Built-in oscillator** (no need for an external clock source).
- Adjustable PWM frequency (40 Hz – 1.6 kHz) for various applications.

- Onboard output enable pin (OE) to quickly disable all outputs.
- Compatible with Python, CircuitPython, and Arduino libraries for easy integration.

Physical Specifications

- Dimensions: 62.5mm × 25.4mm × 3mm
- Weight: 9g

Electrical Characteristics

- Operating Voltage: 3.3V (logic), 5V–6V (servo power)
- Current per Output Pin: Up to 25mA
- Maximum PWM Frequency: 1.6 kHz

Usage Recommendations

- Use a separate 5V–6V power supply for servos to prevent microcontroller power issues.
- If powering multiple servos, add a 470µF or higher capacitor across V+ and GND to stabilize voltage.
- For high-current servos, ensure adequate ventilation or heat dissipation to prevent overheating.

Documentation and Support

- Community Support available on Adafruit forums, GitHub discussions, and Raspberry Pi/Arduino developer communities.

3. **MG995 Servo Motor**

Introduction

The MG995 is a high-torque metal gear servo motor widely used in robotics, RC vehicles, and automation projects. It features dual ball bearings and metal gears, ensuring long-lasting performance and resistance to wear.

Features

- High-torque capability of up to 10 kg·cm torque at 6V.
- Fast response time of 0.16s/60° at 6V, ensuring quick actuation.
- Metal gears for improved durability and longevity.
- Dual ball bearing design for smooth and stable operation.
- Pre-installed 30cm wire with a 3-pin 'S' type female header, compatible with major brands like Futaba, JR, GWS, and Spektrum.

Physical Specifications

- Dimensions: 40.7mm × 19.7mm × 42.9mm
- Weight: 55g

Electrical Characteristics

- Operating Voltage: 4.8V – 7.2V
- Stall Torque: 8.5 kg·cm at 4.8V, 10 kg·cm at 6V
- Dead Band Width: 5 µs

Usage Recommendations

- Secure using servo brackets or custom mounts for stable operation.
- Avoid over-tightening the horn to prevent gear damage.
- For high-load applications, consider heat dissipation or servo cooling fans.

Documentation and Support

- Community Support available on robotics forums, GitHub, and Arduino/Raspberry Pi discussion groups.

4. **FSR402 Force Sensor**

Introduction

The FSR 402 is a round Force Sensing Resistor (FSR) designed to detect changes in force on its surface.

Features

- Detects actuation forces as low as 0.1N, with a range up to 10N
- ±2% repeatability with a single part, ±6% part-to-part repeatability
- Continuous force resolution
- Capable of withstanding up to 10 million actuations
- Operating Temperature: -30°C to +70°C

Physical Specifications

- Dimensions: 18.3mm x 0.55mm
- Weight: 1g

Electrical Characteristics

- Stand-Off Resistance: Exceeds 10MOhm
- Hysteresis: Approximately +10%
- Long-Term Drift: Less than 5%

Usage Recommendations

- Avoid exposing the sensor to temperatures outside of the recommended temperature range.
- Avoid sharp bends near the active sensing area, as this can lead to cracks or separation in the conductive layers.
- In environments with high humidity, consider additional encapsulation or protective housing to reduce moisture exposure.

Documentation and Support

- Interlink Electronics provides detailed support and an integration guide to help users set up the FSR 402.
- For technical assistance or purchasing inquiries, users can reach out via the support email at Interlink Electronics.

5. **MCP3008 ADC**

Introduction

The MCP3008 is a 10-bit analog-to-digital converter (ADC) that enables microcontrollers to interface with analog sensors and devices. It can operate on a single power supply and features eight single-ended input channels or four differential input pairs, offering flexibility in signal measurement.

Features

- 10-bit resolution ADC with ±1 LSB accuracy
- Supports eight single-ended inputs or four differential input pairs
- Maximum sampling rate of 200 ksps at 5V
- SPI-compatible serial interface for efficient data transmission
- On-chip sample and hold circuit for stable signal processing
- Low power consumption, making it suitable for battery-operated devices

Physical Specifications

- Dimensions: 19.3mm × 6.35mm × 3.3mm
- Pin Count: 16 pins

Electrical Characteristics

- Operating Voltage: 2.7V – 5.5V
- Maximum Sampling Rate: 200 ksps (at 5V), 75 ksps (at 2.7V)
- Current Consumption (Active Mode): 500 µA max at 5V

Usage Recommendations

- Use a stable reference voltage to maintain accurate ADC readings.
- Keep SPI communication speed within recommended limits to ensure reliable data transfer.
- Minimize noise by using proper grounding and shielding, especially when working with sensitive analog signals.

Documentation and Support

- Community Support available on Microchip forums, GitHub repositories, and Raspberry Pi/Arduino developer communities